

Homework #1: AR(2) Process Analysis

1. Derivation of Autocovariance Functions

Given a stationary AR(2) process [4]:

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t \quad (1)$$

where $E(x_t) = 0$ and $w_t \sim WN(0, \sigma_w^2)$ with $\sigma_w^2 = 1$.

(a) Calculate $\gamma_x(0)$ in terms of $\gamma_x(1)$ and ϕ_1

To derive this, we multiply the AR(2) equation by x_{t-1} and take the expectation $E[\cdot]$ [1]:

$$\begin{aligned} E(x_t x_{t-1}) &= \phi_1 E(x_{t-1} x_{t-1}) + \phi_2 E(x_{t-2} x_{t-1}) + E(w_t x_{t-1}) \\ \gamma_x(1) &= \phi_1 \gamma_x(0) + \phi_2 \gamma_x(1) + 0 \end{aligned}$$

Isolating $\gamma_x(0)$:

$$\begin{aligned} \phi_1 \gamma_x(0) &= \gamma_x(1) - \phi_2 \gamma_x(1) \\ \gamma_x(0) &= \frac{(1 - \phi_2)}{\phi_1} \gamma_x(1) \end{aligned}$$

(b) Calculate $\gamma_x(1)$ in terms of $\gamma_x(0)$, ϕ_1 , ϕ_2

Using the Yule-Walker relationship derived in step (a) [3]:

$$\begin{aligned} \gamma_x(1)(1 - \phi_2) &= \phi_1 \gamma_x(0) \\ \gamma_x(1) &= \frac{\phi_1}{1 - \phi_2} \gamma_x(0) \end{aligned}$$

(c) General recursive formula for $\gamma_x(h)$

For $h = 2, 3, 4, \dots$, the autocovariance follows the homogeneous difference equation [1]:

$$\gamma_x(h) = \phi_1 \gamma_x(h-1) + \phi_2 \gamma_x(h-2) \quad (2)$$

2. Autocorrelation Coefficients $\rho_x(n)$

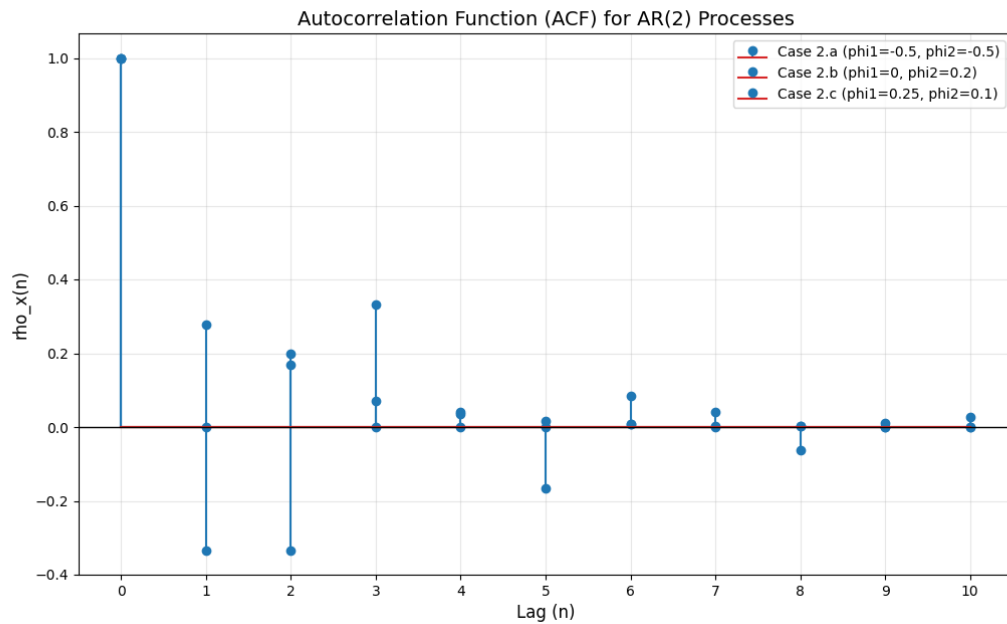
We calculate $\rho_x(n) = \frac{\gamma_x(n)}{\gamma_x(0)}$ using the initial conditions $\rho_x(0) = 1$ and $\rho_x(1) = \frac{\phi_1}{1-\phi_2}$ [2].

n	2.a ($\phi_1 = -0.5, \phi_2 = -0.5$)	2.b ($\phi_1 = 0, \phi_2 = 0.2$)	2.c ($\phi_1 = 0.25, \phi_2 = 0.1$)
0	1.00	1.00	1.00
1	-0.33	0.00	0.28
2	-0.33	0.20	0.17
3	0.33	0.00	0.06
4	0.00	0.04	0.03
5	-0.17	0.00	0.01

Table 1: Calculated values for $\rho_x(n)$.

3. Visualization

The following code computes the values for all three cases and plots the ACF.



```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def calculate_rho(phi1, phi2, lags=10):
5     rho = np.zeros(lags + 1)
6     rho[0] = 1.0
7     rho[1] = phi1 / (1 - phi2)
8     for n in range(2, lags + 1):
9         rho[n] = phi1 * rho[n-1] + phi2 * rho[n-2]
10    return rho
11
12 # Problem cases
13 cases = {"2.a": (-0.5, -0.5), "2.b": (0, 0.2), "2.c": (0.25, 0.1)}
14 x = np.arange(11)

```

```
15
16 plt.figure(figsize=(10, 5))
17 for label, (p1, p2) in cases.items():
18     plt.stem(x, calculate_rho(p1, p2), label=f'Case {label}')
19
20 plt.title("Autocorrelation Function (ACF) - rho_x(n)")
21 plt.xlabel("Lag (n)")
22 plt.ylabel("rho")
23 plt.legend()
24 plt.grid(True)
25 plt.show()
```

4. Analysis of AR Order Estimation

Question: Can we use the ACF ($\rho_x(n)$) plot to estimate the order p of an AR process?

Answer: No, we cannot accurately estimate the order p using only the ACF.

Reasoning: According to the theory [3], the ACF of an AR process **tails off** (decays exponentially or as a damped sine wave) and does not cut off at a specific lag. To identify the order p , we must use the **Partial Autocorrelation Function (PACF)**, which **cuts off** to zero for all lags $h > p$.

References

- [1] ARMA Model Basics, 2102-503 Lecture Slides.
- [2] ARMA Estimator (Forecasting), 2102-503 Lecture Slides.
- [3] Yule-Walker Estimation and PACF, 2102-503 Lecture Slides.
- [4] Homework Assignment #1 Description.